

Toulik Maitra

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Summary

PhD candidate in Chemical Engineering at UC Davis. I build simulation pipelines that go from atomic structure to device-level behavior, with a focus on organic semiconductors and polymer systems. My work spans DFT-based electronic structure, reactive molecular dynamics, and multiscale frameworks, validated against neutron scattering and device measurements at ORNL and the Max Planck Institute. I pick up new methods quickly, work closely with experimentalists, and translate simulation outputs into something engineers can actually use.

Technical Skills

- **Simulation:** DFT (VASP, ORCA, Gaussian 16, DFTB+), molecular dynamics (LAMMPS, GROMACS), reactive MD (ReaxFF-compatible), FEA (ANSYS); process-structure-performance modeling across organic and inorganic material systems
- **Surface & Process Chemistry:** Thin-film growth, defect analysis, thermodynamics and kinetics of surface reactions, atomic-precision characterization; experience correlating computational predictions with experimental data
- **Programming:** Python (NumPy, pandas, scipy, scikit-learn, PyTorch), C/C++, MATLAB; Linux/HPC workflows, Bash scripting, Git; high-throughput pipelines for large-scale simulation data integration and property prediction
- **Experimental Validation:** Neutron scattering at ORNL (SNS/VISION spectrometer); end-to-end workflows from molecular design through fabricated device characterization

Education

University of California, Davis | Ph.D., Chemical Engineering Sep 2022 – Jun 2027

- GPA 4.0/4.0. Research: structure-property-performance correlations in organic semiconductors using multiphysics simulation; built pipelines connecting structural disorder, charge transport, and device performance. Advisor: Adam Moule.
- *Awards:* Graduate Research Award Fellowship (2025); UCD Graduate Studies Fellowship (2025); CHMS Symposium Outstanding Poster (2023).

University of Calcutta | B.Tech., Chemical Engineering Aug 2016 – Aug 2020

- GPA 8.36/10.0. *Awards:* Best Paper (IIT Kharagpur), Most Innovative Idea in Nanoscience (IIT Bombay), Top 10 Global Entrepreneurship Event (IIT Kharagpur).

Professional Experience

Visiting Fellow, Molecular Electronics Group Jul 2025 – Present
Max Planck Institute for Polymer Research, Mainz, Germany. Advisors: Denis Andrienko & Paul Blom

- Improved optical and electronic property prediction accuracy by 20%, validated against fabricated device measurements, by developing excited-state MD workflows that couple polymer chain conformation and crystal morphology to device-level performance in TADF-based OLEDs.
- Reduced force-field parameter error by 60%, benchmarked against quantum-mechanical reference calculations, by building new polarizability parameterization protocols using the Jensen model in VOTCA; the functionality is now part of the production codebase used by the broader research community.
- Increased candidate polymer screening throughput by 30%, measured by structures evaluated per week, by redesigning simulation pipelines for parallel HPC execution and automating extraction of mechanical and optical property metrics.

Graduate Student Researcher Jan 2023 – Present
University of California, Davis

- Cut material property prediction error by 40% relative to prior models, validated against neutron scattering experiments at ORNL (SNS/VISION spectrometer), by building multiscale frameworks that connect polymer microstructural disorder to charge transport, optical response, and mechanical compliance in organic semiconductor systems.
- Built high-throughput simulation pipelines integrating MD, graph-theoretic analysis, and statistical ML across 500+ structural configurations; extracted design rules for intermolecular packing and microstructure optimization, published in *Chem. Mater.*

- Developed a voxelization-based 3D analysis framework for quantifying spatial heterogeneity in molecular packing and tracking how it propagates to device-level property variation; method is under review for publication.

Undergraduate Research Assistant
University of Calcutta, Kolkata, India

Feb 2018 – Dec 2021

- Modeled optical absorption and charge transport in hybrid nanostructures, building methods that correlated atomic structure, processing conditions, and device performance in photoelectrochemical and display device contexts.

Business Development Intern
TATA Autocomp Systems Ltd, Pune, India

May 2019 – Jul 2019

- Reduced component costs by \$100K, validated against procurement baselines, by applying FEA results to cost-performance tradeoff analysis under manufacturing constraints.

Leadership

- **E-Search Mentor:** Supervised 3 undergraduates building multiphysics models of disordered polymer systems; guided them through simulation workflows, validation, and ML-based analysis.
- **Co-Founder, eCell University of Calcutta:** Organized 4 workshops on interdisciplinary prototyping; built a cross-functional team that ran projects across departments.
- **Teaching Assistant, UC Davis** (Apr 2023 – Mar 2025): Taught 7 courses averaging 150+ students each (1,000+ total); developed coursework in thermodynamics, transport phenomena, and materials processing.

Selected Publications

- T. Maitra *et al.* A Voxelization Approach for Correlating MD with INS to Study Local and Global Structural Disorder. *Under review.*
- J. Gong, ..., T. Maitra *et al.* Aggregation Enhanced Singlet Emission in a Purely Organic Emitter with TADF. *Under review.*
- F. Maleki, T. Maitra *et al.* Extensive model validation enables quantitative prediction of complex structural and electronic properties. *Under review.*
- F. Maleki, ..., T. Maitra *et al.* Design Rules to Optimize Intermolecular and Long-Range Packing of Organic Semiconductor Crystals. *Chem. Mater.* **36**(9), 4794–4805, 2024.
- K. Rana, T. Maitra *et al.* Modeling and simulation of spinning basket membrane module in protein recovery. *J. Water Process Eng.* **42**, 102135, 2021. [*and 5 additional peer-reviewed publications*]